

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE-AI	Date:	

Code of Conduct

I P.KALYANI (192472199) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. KALYANI

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2

First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis

Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.

Q3

Annexure A – Industry Problem Solving Task

1. Semantic Representation in Customer Support Chatbots

1.1 Analysis of Action–Object Relationships

Semantic representation converts a natural-language customer query into a structured meaning representation that can be understood by a chatbot.

Customer Query	Semantic Representation	Action	Object	Meaning
Activate international roaming for my number.	ACTIVATE(Roaming, Customer)	ACTIVATE	Roaming	Customer wants roaming activated
Deactivate caller tune service.	DEACTIVATE(CallerTune, Customer)	DEACTIVATE	Caller Tune	Customer wants caller tune deactivated
Check my data balance.	QUERY(DataBalance, Customer)	QUERY	Data Balance	Customer wants to check remaining data
Enable 5G service.	ACTIVATE(5GService, Customer)	ACTIVATE	5G Service	Customer wants 5G activated

The representations follow the general relationship:

Action → Service/Object → Customer

For example:

ACTIVATE(Roaming, Customer)

means that the customer is requesting activation of the roaming service.

Semantic representation is important because the chatbot must convert natural-language requests into structured actions that can be connected to telecom backend services.

1.2 Identification of Semantic Interpretation Error

The chatbot logs show:

Query ID	Actual User Intent	Predicted Intent	Result
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Activate Caller Tune	Incorrect
Q3	Check Data Balance	Query Data Balance	Correct
Q4	Enable 5G Service	Activate 5G Service	Correct

Q2 contains the semantic interpretation error.

The correct representation is:

DEACTIVATE(CallerTune, Customer)

However, the chatbot predicts:

ACTIVATE(CallerTune, Customer)

The object is identified correctly, but the action is incorrectly reversed.

This is a serious error because activation and deactivation are opposite operations. If the predicted intent is sent directly to a backend system, the customer's caller tune could be activated instead of cancelled.

1.3 Effect of Meaning Representation on Chatbot Decision Accuracy

The four queries contain:

- Correct predictions = 3
- Incorrect predictions = 1

Therefore,

Although the chatbot achieves 75% accuracy, this level may not be sufficient for a service-automation system because a single semantic error can result in an incorrect telecom operation.

For example:

Correct:

ACTIVATE(Roaming, Customer)

→ Identify activation request

→ Activate roaming

→ Provide confirmation

Incorrect:

ACTIVATE(CallerTune, Customer)

instead of:

DEACTIVATE(CallerTune, Customer)

→ Wrong backend operation

→ Customer dissatisfaction

→ Additional support request

Thus, semantic representation directly influences the accuracy and reliability of chatbot decision-making.

1.4 Recommended Improvements

The telecom company should implement the following improvements:

1. **Intent Classification** – Distinguish activate, deactivate, query, cancel and renew requests.
2. **Context-Aware Semantic Analysis** – Use previous conversation information.
3. **Negation Detection** – Detect terms such as *not*, *deactivate*, *cancel*, *stop*, and *disable*.
4. **Entity Recognition** – Identify services such as roaming, 5G and caller tune.
5. **Domain-Specific Ontology** – Maintain relationships between telecom services and possible actions.
6. **Confidence Scoring** – Assign a confidence value to each predicted intent.
7. **Confirmation Mechanism** – Ask the customer to confirm critical service operations when confidence is low.
8. **Continuous Learning** – Use corrected customer interactions as training data.

For example:

“Do you want me to deactivate your caller tune service?”

can be presented before performing a potentially incorrect operation.

Conclusion

Semantic representation provides a bridge between natural language and chatbot actions. The given data shows 75% prediction accuracy, but Q2 demonstrates the risk of confusing opposite

actions. Therefore, intent classification, contextual analysis, negation detection, entity recognition and confirmation mechanisms should be combined to improve telecom chatbot reliability.

2. First-Order Predicate Calculus for Smart Manufacturing

2.1 Representation of Production Data Using First-Order Predicate Calculus

The production data is:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

The facts can be represented as:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

Predicate Rules

Rule 1:

Therefore:

Producing(M1)

Producing(M2)

Producing(M4)

Rule 2:

This means that if an active machine produces a product, that product is considered available.

Rule 3:

Therefore:

$\neg$ Producing(M3)

2.2 Analysis of Currently Available Products

The exact products currently available cannot be determined from the supplied data because there is no machine-to-product relationship.

The missing information would be represented as:

Produces(M1,Gear)

If:

Produces(M1,Gear)

Active(M1)

then:

Available(Gear)

can be inferred.

Therefore, the correct conclusion is:

Specific available products cannot be determined because the Produces(x,y) relationships are not provided in the dataset.

This is an important data limitation because predicate logic can only infer conclusions from available facts and rules.

2.3 Effect of Maintenance on Gear Production

The given data establishes:

Maintenance(M3)

Therefore, according to the rule:

we obtain:

$\neg$ Producing(M3)

If M3 produces Gear:

Produces(M3,Gear)

then Gear production by M3 is affected by maintenance.

However, the dataset does not explicitly state:

Produces(M3,Gear)

Therefore, it is not logically valid to conclude that overall Gear production is affected.

Conditional Conclusion

If M3 is responsible for producing Gear, its maintenance activity stops Gear production on M3.

This demonstrates the importance of complete production data for reliable industrial reasoning.

2.4 Effectiveness of Predicate Logic in Industrial Decision Support

First-Order Predicate Calculus provides several advantages.

Advantages

1. **Precise representation** – Machine states can be represented formally.
2. **Rule-based inference** – New facts can be derived automatically.
3. **Explainability** – Decisions can be traced back to logical rules.
4. **Maintenance monitoring** – Machines under maintenance can be identified.
5. **Automation** – The same rules can be continuously applied to new production data.

For example:

and:

Limitations

Predicate logic alone does not handle:

- uncertain information
- noisy sensor readings
- probabilistic failures
- predictive maintenance
- machine-learning-based forecasting

Therefore, a practical Industry 4.0 system can combine predicate logic with IoT sensors, databases, machine learning and rule engines.

Conclusion

Predicate calculus provides an explainable and systematic framework for manufacturing decision support. However, the current dataset lacks machine-to-product relationships. Therefore, additional production mapping is required to determine exact product availability and Gear-production impact.

3. Word Sense Disambiguation in E-Commerce Search Engines

3.1 Determination of Correct Word Sense

The search context and clicked results provide sufficient evidence to determine the intended meaning.

User Query	Possible Sense 1	Possible Sense 2	Clicked Result	Correct Sense
Apple accessories	Fruit	Technology Brand	iPhone Charger	Technology Brand
Mouse wireless	Animal	Computer Device	Bluetooth Mouse	Computer Device

Java tutorial	Island	Programming Language	Coding Lessons	Programming Language
Python course	Snake	Programming Language	Software Development Training	Programming Language

The clicked results provide strong evidence that the users are searching for technology-related meanings.

3.2 Semantic Cues Used for Disambiguation

Apple accessories

The word “accessories” suggests a product-related context. The clicked result iPhone Charger confirms the technology-brand meaning.

Mouse wireless

The term “wireless” is strongly associated with electronic devices. The clicked Bluetooth Mouse confirms that the intended meaning is a computer device.

Java tutorial

The term “tutorial” provides an educational context. The clicked result Coding Lessons confirms that Java refers to the programming language.

Python course

The word “course” combined with software development training indicates that Python refers to the programming language.

The major semantic cues are:

- neighboring words
- query intent
- product category
- clicked results
- historical user behavior
- domain context

3.3 Impact of Incorrect Sense Selection

Incorrect Word Sense Disambiguation can produce irrelevant search results.

Example: Apple accessories

If **Apple** is incorrectly interpreted as a fruit, the system may return:

- fruit baskets
- apple recipes
- agricultural products

instead of:

- iPhone chargers
- AirPods accessories
- Apple Watch accessories

Example: Python course

If Python is interpreted as a snake, the system may return:

- reptile information
- snake-care courses
- animal-related products

instead of programming courses.

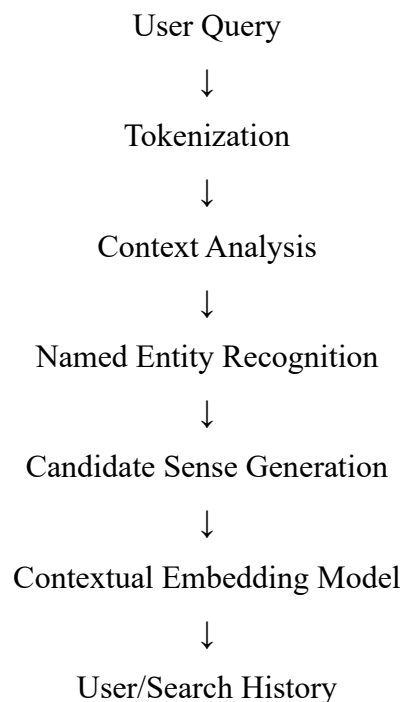
Therefore:

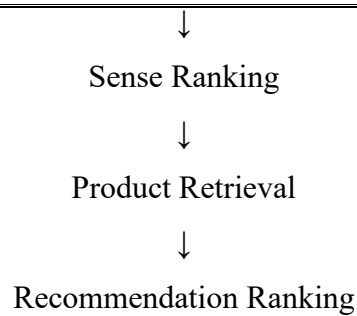
Wrong Sense → Irrelevant Results → Lower Relevance → Fewer Clicks → Lower Conversion → Poor User Satisfaction

For a large e-commerce company, this can directly affect product discovery, sales and customer retention.

3.4 Industrial-Scale WSD Strategy

A hybrid contextual WSD system can be designed as:





Recommended Technologies

1. **Contextual Embeddings** – Understand word meaning according to surrounding words.
2. **Product Ontology** – Represent different meanings and product categories.
3. **Search History** – Use previous user behavior to identify intent.
4. **Click-Through Data** – Use clicked results as feedback.
5. **Feedback Learning** – Learn from clicks, purchases, search reformulation and product views.

For example, an ontology can represent:

Apple

├── Technology Brand
└── Fruit

Mouse

├── Computer Device
└── Animal

Conclusion

The data demonstrates that context is the key factor in resolving ambiguous words. An industrial WSD system combining contextual models, product ontology and user behavior can improve search relevance, recommendations, customer satisfaction and conversion opportunities.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

4.1 Contribution of Syntactic Structures to Semantic Role Assignment

The healthcare system processes:

1. Doctor prescribed medicine to patient.
2. Patient reported severe headache.
3. Nurse monitored patient continuously.
4. Medicine reduced blood pressure.

Syntactic structure helps determine relationships between entities.

For example:

Doctor prescribed medicine to patient.

Structure:

Subject → Verb → Object → Recipient

Therefore:

- Doctor → Agent
- Medicine → Theme/Object
- Patient → Recipient

Similarly:

Patient reported severe headache.

- Patient → Experiencer/Agent
- Headache → Symptom

Thus, syntax provides the structural foundation for semantic-role labeling.

4.2 Evaluation of Assigned Semantic Roles

The given assignments are:

Entity	Assigned Role	Evaluation
Doctor	Agent	Appropriate
Medicine	Instrument	Not appropriate
Patient	Recipient	Appropriate for first sentence
Headache	Symptom	Appropriate

Correction

In:

“Doctor prescribed medicine to patient.”

Medicine should be assigned the role Theme/Object, not Instrument.

An instrument is something used to perform an action.

Example:

“The doctor used a syringe.”

Here:

- Doctor → Agent
- Syringe → Instrument
- Medicine → Theme
- Patient → Recipient

Therefore, Medicine = Instrument is a semantic-role error.

4.3 Potential Errors Due to Incorrect Parsing

Incorrect parsing can result in incorrect clinical interpretation.

Correct interpretation:

Doctor → prescribed → Medicine

Doctor → prescribed → Medicine → to Patient

An incorrect parser could potentially interpret:

Patient → prescribed → Medicine → to Doctor

This changes the complete meaning of the medical statement.

Possible consequences

- Incorrect medical records
- Wrong treatment extraction
- Incorrect patient history
- Incorrect clinical decision support
- Incorrect alerts
- Reduced doctor trust
- Potential patient-safety risks

Therefore, healthcare NLP requires a higher level of semantic accuracy than ordinary information-retrieval applications.

4.4 Recommended Methods for Improvement

1. Dependency Parsing

Dependency parsing identifies grammatical relationships between words.

Example:

Doctor

↓ nsubj

prescribed

↓ dobj

medicine

↓ nmod

patient

This provides detailed relationships between medical entities.

2. Semantic Role Labeling

The system should identify roles such as:

- Agent

- Patient/Theme
- Recipient
- Experiencer
- Instrument
- Location
- Time

3. Medical Named Entity Recognition

The system should recognize:

- Doctor
- Patient
- Medicine
- Disease
- Symptom
- Treatment
- Hospital
- Location

4. Medical Ontology

Domain knowledge should distinguish:

Medicine $\neq$ Instrument

Patient $\neq$ Treatment

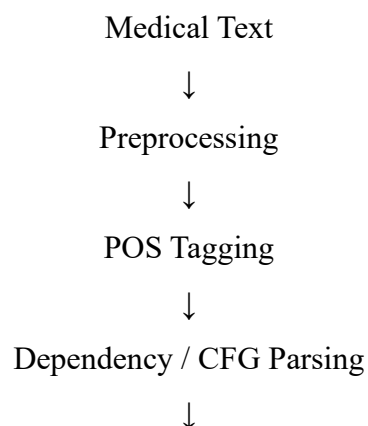
Disease $\neq$ Medication

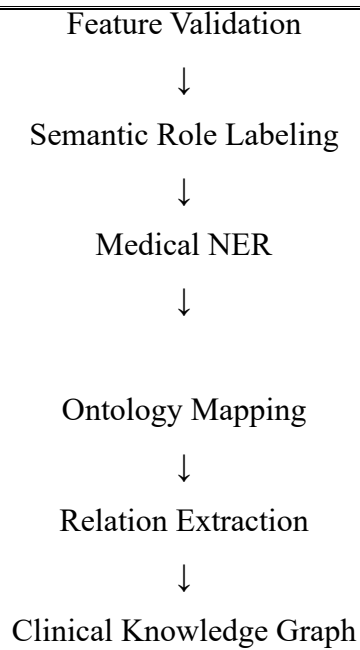
5. Domain-Specific Parsing

Generic NLP parsers may not understand specialized medical terminology. Medical corpora and domain-specific models should therefore be used.

6. Hybrid Syntax-Semantic Architecture

A practical architecture is:





Conclusion

Syntax provides the basic structure, but syntax alone is insufficient for accurate medical interpretation. Combining dependency parsing, semantic role labeling, medical NER, domain ontology and relation extraction can significantly reduce semantic errors and improve the reliability of healthcare NLP systems.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____